

TrES-3b

R-0303

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_180404 · created 2026-07-17T21:11:10+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458212.8858 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.023
Transit depth (Rp/Rs)^2	2.92 +/- 0.79 [%]
Orbital Inclination (inc)	82.16 +/- 1.38
Airmass coefficient 1 (a1)	1.007 +/- 0.011
Airmass coefficient 2 (a2)	-0.0058 +/- 0.0065
Scatter in the residuals of the lightcurve fit is	1.08 %
Best Comparison Star	#1 - [115, 57]
Optimal Aperture	3.12
Optimal Annulus	11.96
Transit Duration (day)	0.055 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.171 ± 0.023 vs 0.16309 (deviation 0.34σ)
Duration fit vs published	0.055 d vs 0.0567 d (deviation 0.11σ)
Mid-time O–C	2.6 min
Depth SNR	7.4
Residual scatter	1.08 %

Light curve

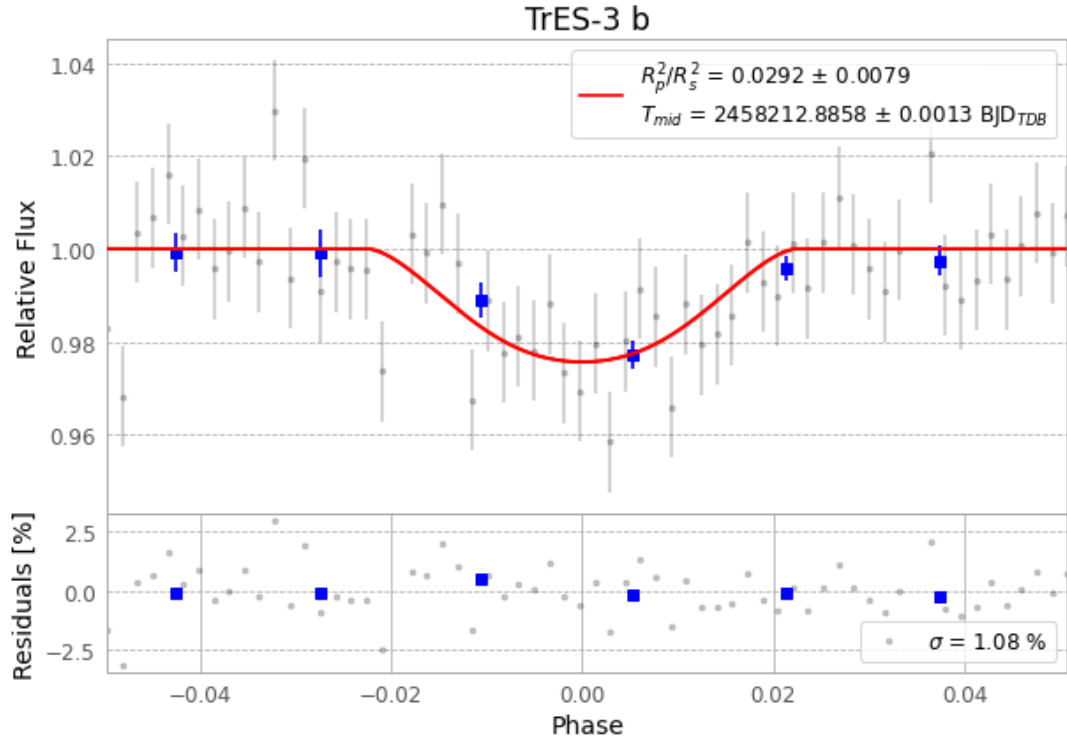


Figure 1 — FinalLightCurve_TrES-3 b_2018-04-04.png (SHA-256 c1aea6adbf750f2a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [416, 179], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8061563f4ca4df69...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 055cf2e

Data provenance

64 FITS frames (42 MB), TRES-3180404073912.FITS → TRES-3180404104812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-180404.log (ac923ae428d7...), FinalLightCurve_TrES-3 b_2018-04-04.png (c1aea6adbf75...), FinalLightCurve_TrES-3 b_2018-04-04.csv (4962ca081acd...)

Compute resources

Wall time 130.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 21:11Z.